

From: Jacob Bevier bevier19jac@icloud.com

Subject: Re: GT referral from Prof. Ng — aerosol spray characterization & VOC-compliant formulation

Date: September 17, 2025 at 1:38:27 PM

To: Weber, Rodney J rweber@eas.gatech.edu

Cc: sarahjbraganini@gmail.com sarahjbraganini@gmail.com

Dear Prof. Weber,

I'm local—in Tyrone, about 15 minutes south of the Atlanta airport. I would be happy to have a conversation in person or virtually, even if you can't help us directly. Just from our email correspondence I have learned a lot about some of these processes that need to happen from you.

If you're able to point me toward the **aerosol experimentalists in EAS** you mentioned, I'd really appreciate the introductions. I'm eager to speak with the right group about the practical formulation and measurement work we discussed.

Thank you again for your time. Talk soon.

Very respectfully,

Jacob Bevier

Co-Founder, Brag & Bev LLC

 760-490-3416 |  bevier19jac@icloud.com

Very respectfully,

Jacob Bevier

7604903416

On Sep 17, 2025, at 1:31 PM, "Weber, Rodney J" <rweber@eas.gatech.edu> wrote:

Where are you located?

There are other faculty in EAS who are aerosol experimentalists, but outside of this I am now aware of anyone at GT who does this.

rW

From: Jacob Bevier <bevier19jac@icloud.com>

Date: Wednesday, September 17, 2025 at 9:49 AM

To: Weber, Rodney J <rweber@eas.gatech.edu>

Cc: sarahjbraganini@gmail.com <sarahjbraganini@gmail.com>

Subject: Re: GT referral from Prof. Ng — aerosol spray characterization & VOC-compliant formulation

Dear Prof. Weber,

Thank you for your thoughtful response and for outlining the considerations on droplet size, solvents, and deposition methods — I'm learning a great deal as I go. To be candid, I am a retired veteran now working as a cybersecurity analyst with a Master's in Cybersecurity/AI, so this field is outside my core expertise. That said, my co-founder and I have invested heavily in this project, and we're determined to align with the right team who has the aerosol science background to help us bridge this gap.

To clarify our need:

- Our current prototype performs like a **deodorizer**. Our challenge is evolving it into a true **cleanser** formulation while staying within **VOC compliance**.
- The system must provide reliable coverage inside a confined space of **30" W × 30" L × 50" H**, with consistent atomization, minimal pooling, and sufficient vertical reach.
- The aerosol portion is only one part of the larger design — it will ultimately be housed within a **mechanical unit** (patent-pending) that ensures can stability and automatic actuation during operation.

We're trying to understand how to take the step from "strong spray performance" to "verified cleaning performance under VOC limits" and to identify the right solvent/propellant/active package balance. I appreciate the tracer suggestion (e.g., fluorescein) as a way to experimentally quantify coverage — that seems very aligned with what we'll need to validate our design.

In short, we're less in need of a broad \$150k/year research program and more in need of **practical formulation and measurement guidance** that gets us from baseline to a VOC-compliant cleanser with reliable performance. Whether that's a consulting arrangement, collaboration, or referral to a group at GT better suited to applied aerosol formulation, I would greatly value your advice.

Thank you again for your time and for sharing your expertise. It's extremely helpful as we navigate this.

Very respectfully,

Jacob Bevier

Co-Founder, Brag & Bev LLC

 760-490-3416 |  bevier19jac@icloud.com

On Sep 16, 2025, at 5:46 PM, "Weber, Rodney J" <rweber@eas.gatech.edu> wrote:

To clarify, if you are looking at some form of \$ support for a research type project with paying GT overhead (more than 60%) and support for a grad student with tuition covered you're looking at minimum \$150k per year.

Or I guess you could try some type of consulting arrangement.

I don't have any idea what kind of facilities you have but maybe this is not that difficult a problem. Most nebulizers, or atomizers, produce around 3 to 10 μm drops. The final size is controlled by how much volatile solvent is in the solution atomized and so eventually in the drop, eg water is the most common. Typically, you atomize an aerosol then dry it (remove the solvent, water in this case) and are left with the non volatile species – this is standard aerosol technology. If you have no solvent that evaporates you have the original drops from the atomizer, so the size range is somewhat set. Not sure how viscosity affects size of atomized drops, could be more related to surface tension. Drops larger than 10 μm fall out quickly and so hard to use effectively.

As for deposition on the part of interest, I would just put a tracer in the solution going through the atomizer and detect it on the part of interest. Eg, a fluorescent tracer that would

be readily visible with a black light or light of specific wavelength. Need to find a compatible tracer for the solvent used (ie, the liquid in the atomizer along with your disinfectant, or whatever it is).

Check out fluoresceine, it is soluble in water and commonly used for these types of applications.

rw

From: Jacob Bevier <bevier19jac@icloud.com>

Date: Tuesday, September 16, 2025 at 3:40 PM

To: Weber, Rodney J <rweber@eas.gatech.edu>

Cc: sarahjbraganini@gmail.com <sarahjbraganini@gmail.com>

Subject: Re: GT referral from Prof. Ng — aerosol spray characterization & VOC-compliant formulation

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Dear Prof. Weber,

Thank you for breaking down the distinctions—very helpful. To clarify:

- Yes, we're interested in **both** the **particle/droplet size distribution (PSD/SMD)** *and* applied measurements of spray behavior in a controlled space: **coverage uniformity on a surface, pooling, and vertical reach**. We see these as complementary: PSD tells us what the spray *is*, while the chamber/surface tests tell us how it *behaves* as a cleanser.
- We **do not yet know** the exact droplet size; quantifying that (and linking it to formulation and hardware choices) is a big part of why we're seeking expert support.

Beyond measurement, we also need **formulation development** help to move from a deodorizer baseline to a **true cleanser** while staying **VOC-compliant**. Concretely, we're hoping to iterate on:

- Solvent/propellant ratios and **viscosity control** for atomization consistency,
- **Surfactant/cleaning package** selection that delivers measurable cleaning action within VOC limits,
- **Actuator/nozzle/can pairing** so the chosen formulation performs reliably across sprays.

If your group can support either (A) the **measurement campaign**, (B) **formulation guidance/iteration** under VOC constraints, or (C) a **joint effort** across both, that would be ideal. If not, any referral within GT (or nearby collaborators) who focuses on **applied aerosol formulation** would be greatly appreciated.

Very respectfully,

Jacob Bevier

Co-Founder, Brag & Bev LLC

📞 760-490-3416 | ✉ bevier19jac@icloud.com

On Sep 16, 2025, at 3:30 PM, "Weber, Rodney J" <rweber@eas.gatech.edu> wrote:

Please clarify. I understand you want a measurement of the droplet size, but what is PSD (particle size distribution?, (if so how is that different from droplet size). You also want some way to determine for your atomizer and "cleanser" (I assume that is some form of chamber) a way to determine the coverage uniformity on a surface within the chamber, pooling of the drops once on the surface and vertical reach of the droplets in the chamber – which are very different measurements from measuring the droplet size distribution.

Is this correct?

Roughly, do you know what the size is of the droplets – that will determine if or how they would be measured.

rW

From: Jacob Bevier <bevier19jac@icloud.com>

Date: Tuesday, September 16, 2025 at 1:13 PM

To: Weber, Rodney J <rweber@eas.gatech.edu>

Subject: GT referral from Prof. Ng — aerosol spray characterization & VOC-compliant formulation

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Dear Professor Weber

Prof. Nga Lee (Sally) Ng suggested I contact you.

I'm Jacob Bevier, co-founder of Brag & Bev LLC. We're advancing a VOC-compliant aerosol cleanser and seeking collaboration on spray characterization (droplet size/PSD, coverage uniformity, pooling, vertical reach) and guidance on formulation within VOC limits plus actuator/nozzle/can optimization.

Status & needs (brief):

- Prototype achieves strong atomization with minimal pooling on a deodorizer baseline.
- Key variables: viscosity, propellant behavior, VOC compliance, repeatability.
- We have funding, can execute an NDA, and can share a concise PowerPoint with test results/images.

If this might fit your lab's expertise, could we schedule a brief call? I'd value your perspective or any referrals you recommend within GT.

Very respectfully,

Jacob Bevier

Co-Founder, Brag & Bev LLC

 760-490-3416 |  bevier19jac@icloud.com